

Spring MVC, Spring Boot & JPA (Hibernate ORM)

Spring MVC Interview Questions

1. What is Spring MVC?
2. Explain the architecture of Spring MVC.
3. What is the role of DispatcherServlet?
4. What is the front controller in Spring MVC?
5. What are the main components of Spring MVC?
6. What is the difference between @Controller and @RestController?
7. What is @RequestMapping used for?
8. How do you handle GET and POST requests in Spring MVC?
9. What is the difference between @GetMapping and @PostMapping?
10. What is a Model in Spring MVC?
11. What is the purpose of ModelAndView?
12. What are @PathVariable and @RequestParam annotations?
13. How can you handle form submission in Spring MVC?
14. How do you perform data validation in Spring MVC?
15. How can you handle exceptions globally in Spring MVC?
16. What is @ExceptionHandler annotation?
17. How do you return JSON data from a controller?
18. How to upload files in Spring MVC?
19. What is the difference between redirect: and forward:?
20. How do you configure Spring MVC without XML (Java-based config)?
21. How to integrate Spring MVC with Hibernate/JPA?
22. How do you manage sessions in Spring MVC?
23. How do you use @SessionAttributes and @ModelAttribute?
24. How do you internationalize a Spring MVC application?
25. How can you secure a Spring MVC application?

Spring Boot Interview Questions

26. What is Spring Boot?
27. What are the advantages of using Spring Boot?
28. What is the difference between Spring and Spring Boot?
29. What is the use of application.properties or application.yml?
30. What are Starters in Spring Boot?
31. What is the purpose of @SpringBootApplication?
32. What are the main components included in @SpringBootApplication?

33. What is auto-configuration in Spring Boot?
 34. How does Spring Boot automatically configure beans?
 35. How do you create a Spring Boot REST API?
 36. What is @RestController?
 37. What is the difference between @Component, @Service, @Repository, and @Controller?
 38. How do you inject dependencies in Spring Boot?
 39. What are the different ways to configure a Spring Boot application?
 40. What is the role of @Value and @ConfigurationProperties?
 41. What is CommandLineRunner?
 42. What is the use of application.properties file?
 43. What is @EnableAutoConfiguration?
 44. How do you exclude specific auto-configuration classes?
 45. How do you create a custom banner in Spring Boot?
 46. How do you connect Spring Boot to a database?
 47. What is the use of spring.datasource.url property?
 48. What is Spring Data JPA?
 49. How do you configure JPA in Spring Boot?
 50. What is the use of spring.jpa.hibernate.ddl-auto?
 51. What is the difference between create, update, and validate in Hibernate DDL-auto?
 52. How do you execute native queries in Spring Data JPA?
 53. How to define custom queries using @Query annotation?
 54. How do you enable pagination and sorting in Spring Data JPA?
 55. How do you handle database transactions in Spring Boot?
 56. What is Actuator in Spring Boot?
 57. What is the difference between @RestControllerAdvice and @ControllerAdvice?
 58. How can you handle exceptions globally in Spring Boot?
 59. How to use Swagger/OpenAPI with Spring Boot?
 60. How to implement validation using @Valid and @NotNull annotations?
 61. How do you manage different profiles in Spring Boot?
 62. How do you secure a REST API using Spring Security?
 63. What is the difference between synchronous and asynchronous REST endpoints?
 64. How do you handle file uploads and downloads in Spring Boot REST?
 65. How do you consume REST APIs using RestTemplate or WebClient?
- JPA & Hibernate ORM Interview Questions
66. What is JPA?
 67. What is the difference between JPA and Hibernate?
 68. What are ORM and its advantages?
 69. What are some ORM tools used in Java?

70. What are entities in JPA?
71. What are annotations used for mapping entities to database tables?
72. What is @Entity annotation?
73. What is the role of @Id and @GeneratedValue?
74. What is the difference between GenerationType.AUTO, IDENTITY, and SEQUENCE?
75. What is @Table annotation used for?
76. What are the different types of entity relationships in JPA?
77. Explain @OneToOne, @OneToMany, and @ManyToMany.
78. What is the difference between fetch = FetchType.LAZY and fetch = FetchType.EAGER?
79. What are Cascade types in JPA?
80. How do you define a composite primary key in JPA?
81. What is the difference between merge() and persist() methods?
82. How can you update or delete an entity?
83. What is JPQL (Java Persistence Query Language)?
84. Difference between JPQL and SQL?
85. How to execute JPQL and native queries in JPA?
86. What is the difference between first-level and second-level caching in Hibernate?
87. What is the default cache level in Hibernate?
88. How do you enable second-level cache?
89. What is optimistic and pessimistic locking in Hibernate?
90. What is dirty checking in Hibernate?
91. What is the N+1 select problem and how to solve it?
92. How to handle transactions in JPA using @Transactional?
93. How can you manage database versioning and schema updates in Spring Boot?
94. What is the role of EntityManager in JPA?
95. How do you manage bi-directional relationships in JPA?
96. Explain the flow: Controller → Service → Repository → Database in Spring Boot.
97. How do you perform CRUD operations using Spring Boot + JPA?
98. How do you validate and save form data from JSP or Thymeleaf in Spring Boot?
99. How do you handle exceptions while saving entities in JPA?
100. How do you optimize performance in Hibernate-based applications?
101. What is Spring Security and why is it used?
102. Explain how authentication and authorization work in Spring Security.
103. What is the difference between AuthenticationManager and UserDetailsService?
104. What is a SecurityContext and SecurityContextHolder?
105. What is the purpose of PasswordEncoder in Spring Security?
106. How do you configure role-based access control in Spring Security?
107. What are filters in Spring Security and how are they used?
108. How do you secure REST APIs using JWT in Spring Security?
109. What are the common annotations used in Spring Security?
110. How do you configure custom login and logout in Spring Security?
111. What is CSRF and how is it handled in Spring Security?

112. How do you implement method-level security in Spring?
113. What is the use of `@EnableWebSecurity` annotation?
114. How do you integrate OAuth2 in Spring Boot application?
115. How do you handle stateless authentication in Spring Security?
116. What are common ways to secure REST endpoints?
117. Suppose you are developing an Employee Management Microservice — how will you structure your microservices (Employee, Department, Project)?
118. In your project, how do different microservices communicate with each other — using REST, Feign Client, or Message Broker?
119. You have multiple microservices accessing different databases — how do you maintain data consistency across them?
120. How will you handle a use case where you need data from multiple microservices in a single API response?
121. Your authentication service issues JWT tokens — how will you secure internal microservices using that token?
122. What happens if your User Service is down but your Order Service still depends on it? How will you make your system resilient?
123. How do you handle configuration across multiple microservices (e.g., Spring Cloud Config Server)?
124. In a production setup, if one microservice goes down, how do you detect and recover automatically?
125. How would you implement service discovery and load balancing between microservices?
126. How would you design your system to support zero downtime deployment?
127. You have a relationship between Employee and Department. How do you design it using JPA?
128. You notice performance issues due to N+1 select problem in Hibernate — how do you resolve it?
129. How do you handle large data fetching (say 10,000 records) without affecting performance?
130. In your project, an entity has soft delete requirement — how do you implement it in JPA?
131. You need to fetch data from multiple tables with dynamic filters — how do you implement it using Criteria API or Specification?
132. How do you handle lazy loading exceptions when returning entity data via REST APIs?
133. Suppose you have an entity Order with `@OneToMany` relation to OrderItem. How will you design efficient queries?
134. How would you handle transaction rollback when one DB operation fails but others succeed?
135. What is your approach to handle batch inserts or updates in JPA?
136. You need to migrate from one DB to another — how do you manage schema migration (Flyway/Liquibase)?
137. Suppose OrderService needs customer details from CustomerService — how do you implement this using Feign Client?
138. If Feign Client call fails, how do you ensure system doesn't crash (use Resilience4j /

Circuit Breaker)?

139. How do you propagate JWT token between microservices in secured REST calls?
140. You have API Gateway in front of microservices — how do you validate JWT only once at gateway level?
141. How do you handle Cross-Origin Resource Sharing (CORS) in your Spring Boot microservices?
142. If multiple microservices access the same database, how do you maintain transactional consistency?
143. How do you ensure idempotency for API requests like payment or booking creation?
144. How would you implement global exception handling across all microservices?
145. If you need centralized logging from all microservices, how do you achieve it (e.g., ELK Stack)?
146. How would you manage and monitor microservices health in production?
147. What are the main features introduced in Java 8?
148. What are functional interfaces in Java?
149. What is a lambda expression and how is it used?
150. What is the Stream API in Java 8?
151. What is the difference between intermediate and terminal operations in streams?
152. How do you use Optional in Java 8?
153. What is the purpose of the default and static methods in interfaces?
154. What is the difference between map() and flatMap()?
155. What are new features introduced in Java 11 and Java 17 LTS?
156. How do you use var keyword introduced in JDK 10?
157. What is the difference between record and class in Java 16?
158. How do you use switch expressions in Java 14+?
159. What are sealed classes in Java 17?
160. What is the difference between LocalDate, LocalDateTime, and ZonedDateTime?
161. How do you use the Predicate, Function, and Consumer interfaces?
162. What is method reference in Java?
163. What are the advantages of Stream API over loops?
164. How do you parallelize stream operations?
165. What is the difference between Optional.of() and Optional.ofNullable()?
166. How do you filter and collect stream data into a Map?
167. Your REST API is returning data very slowly — how would you debug and fix it?
168. A Hibernate query is triggering 50 SQL queries for one API — what's wrong?
169. How do you handle circular dependencies between microservices?
170. How would you design an Employee-Department microservice architecture?
171. How do you handle lazy loading exceptions in REST responses?
172. Your DB connection pool is exhausted — what steps will you take?
173. How do you monitor and analyze memory leaks in a Spring Boot app?
174. How do you migrate a monolithic application into microservices?

175. How do you ensure fault tolerance between microservices?
176. How do you handle data synchronization across multiple services?
177. What is your deployment process for microservices (CI/CD)?
178. How do you implement role-based access control (RBAC) using Spring Security?
179. How do you implement request logging for debugging?
180. How do you cache frequently accessed data using Spring Cache or Redis?
181. How do you secure sensitive configurations in production (e.g., DB credentials)?
182. How do you handle version upgrades (JDK 8 → 17 → 23) in your projects?
183. How do you reduce startup time of a Spring Boot microservice?
184. How do you handle database migrations between environments (dev → prod)?
185. How do you test integration between microservices locally?
186. How do you use Docker for microservice deployment?
187. How do you handle API backward compatibility?
188. How do you detect and fix Hibernate performance bottlenecks?
189. What monitoring tools do you use for production (Prometheus, Grafana, ELK)?
190. How do you implement rate limiting in APIs?
191. Describe a recent project where you optimized microservice performance.
192. What are ACID properties in a relational database?
193. What is normalization? Explain 1NF, 2NF, 3NF.
194. What is the difference between INNER JOIN and LEFT JOIN?
195. How do you optimize a slow SQL query?
196. What are indexes? When should you use or avoid them?
197. What is a foreign key constraint and why is it important?
198. How do you manage database migrations in Spring Boot (Flyway/Liquibase)?
199. What's the difference between clustered and non-clustered indexes?
200. How do you handle transactions across multiple tables?
201. What is the difference between CHAR and VARCHAR data types?
202. How do you prevent SQL injection in Spring Boot applications?
203. What are stored procedures and how do you call them from JPA?
204. How do you implement soft deletes in a table?
205. How do you handle connection pooling in Spring Boot JPA?
206. How do you configure multiple data sources in a single Spring Boot project?
207. What is REST architecture and its core principles?
208. What is an idempotent operation in REST?
209. How do you handle exception responses in REST APIs?
210. What are the best practices for designing RESTful APIs?
211. What are HTTP status codes and give some common examples.
212. How do you handle validation in REST APIs?
213. What is the difference between path variable and request parameter?

214. How do you secure REST APIs?
215. What is JWT and how is it used for authentication?
216. How do you handle file uploads in a REST API?
217. What are HATEOAS principles?
218. How do you handle pagination and sorting in REST APIs?
219. How do you test REST APIs using Postman or Swagger?
220. What is Swagger/OpenAPI and how do you integrate it with Spring Boot?
221. How do you version REST APIs?
222. How do you handle request and response logging in REST APIs?
223. How do you return consistent API responses?
224. What is the difference between synchronous and asynchronous REST APIs?
225. How do you handle large data responses in REST APIs?
226. How do you consume a REST API in Spring Boot using RestTemplate or WebClient?
227. What is the difference between RestTemplate and WebClient?
228. What is content negotiation in REST APIs?
229. How do you handle rate limiting and throttling in REST APIs?
230. What is a microservice architecture?
231. What are the advantages and disadvantages of microservices?
232. How is microservice architecture different from monolithic architecture?
233. What are the main components of a microservice ecosystem?
234. How do microservices communicate with each other?
235. What is the role of Spring Cloud in microservices?
236. What is service discovery and how is it implemented using Eureka or Consul?
237. How do you implement API Gateway in microservices?
238. What is the role of Spring Cloud Gateway or Zuul?
239. How do you secure communication between microservices?
240. How does JWT token work across multiple microservices?
241. What is configuration management and why is Spring Cloud Config used?
242. How do you handle distributed tracing in microservices?
243. What tools are used for tracing (Zipkin, Sleuth, Jaeger)?
244. What is load balancing in microservices and how is it achieved?
245. What is the role of Feign Client in microservices?
246. How do you handle circuit breakers using Resilience4j or Hystrix?
247. What is service registry and discovery pattern?
248. How do you handle centralized logging in microservices?
249. How do you manage database per service in microservice architecture?
250. How do you achieve transaction management across microservices?
251. What is eventual consistency and how do you achieve it?
252. How do you version microservices APIs?

- 253. How do you test and deploy microservices independently?
- 254. What is a sidecar pattern in microservices?
- 255. How do you containerize and deploy microservices using Docker?
- 256. How do you orchestrate microservices using Kubernetes?
- 257. How do you handle inter-service communication failures?
- 258. What is a distributed cache and how is it used in microservices?
- 259. How do you handle fault tolerance and resilience?
- 260. What is API Gateway throttling?
- 261. How do you secure microservices using OAuth2?
- 262. How do you handle synchronous vs asynchronous communication between services?
- 263. How do you share common libraries across multiple microservices?
- 264. What is a saga pattern in microservices?
- 265. How do you handle schema evolution in databases across microservices?
- 266. What monitoring tools do you use for microservices (Prometheus, Grafana, ELK)?
- 267. How do you handle deployment rollbacks in microservice architecture?
- 268. How do you use Circuit Breaker with Feign Client?
- 269. What are some real challenges you faced working with microservices in production?